



An improved MOEA/D algorithm with an adaptive evolutionary strategy



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ARTICLE INFO

Article history:

Received 11 November 2019

Received in revised form 14 April 2020

Accepted 20 May 2020

Available online 7 June 2020

Keywords:

MOEA/D

Multi-objective optimization

Adaptive evolutionary strategy

ABSTRACT

The Multi-objective Evolutionary Algorithm Based on Decomposition (MOEA/D) overcomes the limitation of evolutionary algorithm based on a Pareto dominant relationship in dealing with the problem of super multi-objective optimization and has wide application prospects, but there are also some problems, such as the lack of diversity and slow convergence speed in the later-stage evolution species. This article specifically conducts a systematic study on the population diversity of the MOEA/D algorithm and proposes three improvements: firstly, the evolutionary strategy of competition between SBX and DE operator is adopted to overcome the problem of the species diversity degradation of a single operator; secondly, an adaptive adjusting strategy of modulation probability is introduced to promote the variability of later-stage evolution species; finally, a method of double-faced mirror theory boundary optimization is used to prevent species aggregating at the boundary. The research shows that the above three improvement measures can effectively improve the population diversity of the MOEA/D algorithm. On the basis of this research, an improved MOEA/D algorithm with adaptive evolution strategy (AES-MOEA/D) is proposed. Simulation experiment indicators show that the convergence and comprehensive performance of the AES-MOEA/D algorithm are better than that of the basic MOEA/D algorithm and the other four comparison algorithms, which shows that the research on the maintenance of the diversity of the MOEA/D algorithm population helps improve the comprehensive performance of the MOEA/D algorithm.

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1. Introduction

Multi-objective optimization problems (MOPs) play an important role in scientific research, production practice and engineering applications, and the research on MOP issues has important theoretical and academic value. MOP optimization can obtain a series of solution sets, and the performance evaluation and decision selection of these solution sets are the problems faced by multi-objective optimization research.

Compared with single-objective optimization, a MOP's goal is no longer a single optimal solution but instead a mutually constrained solution set [1], i.e., the improvement of one optimization objective may be accompanied by the deterioration of

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other optimization goals. Thus, the academic community generally uses the Pareto solution set [2] to represent a solution set of various target trade-offs in multi-objective optimization, while the projection of the Pareto optimal solution set in the target domain is called the Pareto Front (PF) [3].

Simple MOPs can be solved by mathematical programming, but this method will become extremely complicated when faced with high-dimensional MOPs [4]. To this end, evolutionary algorithms with heuristic search features are introduced to solve multi-objective optimization problems [5]. Early multi-objective evolutionary algorithms, such as SPEA [6], SPEA-II [7], NSGA-II [8], PESA [9], PESA-II [10] and so on, are evolutionary algorithms based on Pareto's dominant relationship. Although these evolutionary algorithms have achieved remarkable results in the field of multi-objective optimization, they meet challenges when faced with the optimization of many-objective problems [11] (objective numbers equal to or greater than 4). In 2007, Zhang [12] and others proposed a multi-objective evolutionary algorithm (MOEA/D) based on decomposition, which decomposes the multi-objective optimization problem into a set of single-objective optimization problems and then uses the evolutionary algorithm to solve these decomposition problems at the same time. The MOEA/D algorithm also introduces neighbor relationships to share evolutionary information. In addition, the MOEA/D algorithm has been directly applied to the problem of many-objective optimization, and it overcomes the limitations of the evolutionary algorithm based on the Pareto dominant relationship. Therefore, the MOEA/D algorithm framework has become a popular research topic in recent years.

The MOEA/D algorithm has undoubtedly gained great success in the field of multi-objective optimization, but there are still some problems, such as the lack of diversity and slow convergence speed in the later-stage evolution species. To this end, scholars have put forward a variety of MOEA/D algorithm improvement projects, with the following main aspects of improvement: weight vector improvement [13], decomposition method improvement [14,15], evolutionary operator improvement [16–18], matching selection improvement [19–21], and replacement selection improvement [22,23]. However, these studies have done a little to solve the lack of diversity of MOEA/D algorithm species and lack systematic solutions to the problem. In view of the lack of species diversity of the MOEA/D algorithm, this paper makes three kinds of improvement to the algorithm.

First, a cross-evolution strategy of competition between simulated binary crossover (SBX) [24] and differential evolution (DE) [25,26] is used. The SBX algorithm has the characteristic of fast convergence but can easily fall into a local optimal solution, while the DE algorithm has strong global search ability but slow convergence. Combining the two algorithms not only balances the convergence speed and global search capabilities of the SBX algorithm and the DE algorithm but also overcomes the problem of lack of population diversity in the late evolution.

Second, a mutation strategy with adaptive mutation probability is used instead of the fixed mutation probability in the MOEA/D algorithm. In evolutionary algorithms, setting the mutation probability too large will increase the blindness of the search, which will reduce the algorithm's convergence speed. Setting the mutation probability too small will make the diversity of the population lack in the later evolution. Therefore, it is difficult to set a fixed mutation probability to adapt to the entire evolutionary process of the evolutionary algorithm, and adaptive mutation probability is a trend of improvement of the evolutionary algorithm [27,28]. According to the characteristic of the MOEA/D algorithm, this paper proposes a mutation strategy whose mutation probability is increased in negative exponential in the evolution process. The mutation probability is set small in the early evolution to obtain a faster convergence rate and then it will become larger, approaching a fixed value, to obtain more diverse subpopulations in the later evolution. If evolution falls into a stagnation state, the algorithm will temporarily increase the mutation probability to seek to break through the stagnation state as quickly as possible.

Finally, the double-faced mirror theory is used to optimize the boundary processing problem in the MOEA/D algorithm. The study found that, during the evolution of the MOEA/D algorithm, its approximate Pareto optimal solution set will gather near the boundary area, while other areas are sparsely distributed, which also affects the diversity of its population. This paper introduces the double-faced mirror theory to deal with the boundary problem of the MOEA/D algorithm. This method can make cross-boundary individuals be uniformly distributed in the definitional domain after the boundary processing and can effectively solve the problem of the approximate Pareto optimal solution set gathering at the boundary.

After verifying the effectiveness of the above three improvement measures, this paper proposes an improved MOEA/D algorithm with an Adaptive Evolutionary Strategy (AES-MOEA/D) and then conducts numerous comparative simulation experiments. Simulation experiments show that the AES-MOEA/D algorithm has improved convergence performance and comprehensive indicators compared with the MOEA/D algorithm and the SPEAII, NSGA-II, and PESA-II algorithms.

2. Related works

2.1. Multi-objective optimization problems and related concepts.

In the case of minimizing multi-objective issues, the MOP can be described as:

Decision variable $X = (x_1, x_2, \dots, x_n)$ meets constraints:

$$\begin{cases} g_i(X) \geq 0 & i = 1, 2, \dots, k \\ h_i(X) = 0 & i = 1, 2, \dots, k \end{cases} \quad (1)$$

If the number of objective functions is m , then the optimization target is represented as:

$$F(X) = (f_1(X), f_2(X), \dots, f_n(X))^T \quad (2)$$

Then, following the constraints, the MOP is to find a solution set $X^* = (x_1^*, x_2^*, \dots, x_n^*)$ in a decision space to make $F(X^*)$ minimum. On this basis, the following definitions are given:

(1) Feasible solution set X_f : a collection of decision variables x that meet constraint (2), i.e., $X_f = \{x \in X | g(x) \geq 0, h(x) = 0\}$

(2) Pareto domination:

If p and q are any of the two different individuals in the evolutionary group, and if ① all of p 's subobjectives are not inferior than q , i.e., $f_k(p) \leq f_k(q) (k = 1, 2, \dots, n)$ and ② p has at least one subobjective better than q , i.e., p dominates q , then p is dominant, and q is dominated. [29]

(3) Pareto optimal solution:

If $x^* \in X_f$, and if there is no other solution $x'^* \in X_f$ to make $f_i(x'^*) \leq f_i(x^*) (i = 1, 2, \dots, m)$, and at least there is one strict in equation, then x^* is the Pareto optimal solution of this MOP. [30]

(4) Pareto Optimal Solution Set:

The collection of all Pareto optimal solutions for MOPs is the Pareto Set (PS).

(5) Pareto Front:

The projection of the Pareto Set in the target space is the Pareto Front (PF), i.e., $PF = \{F(x) | x \in PS\}$.

2.2. Introduction of the MOEA/D algorithm

The MOEA/D algorithm uses aggregation functions to decompose MOPs into N sub-problems and optimize them at the same time. This algorithm divides each sub-problem by calculating the Euclidean distance [31] among the weight vectors. And each sub problem evolves collaboratively through sharing their neighborhood; the MOEA/D algorithm process is as follows:

Step 1: Initialization

- (1) Initialize the weight vector $\lambda = \{\lambda_1, \lambda_2, \dots, \lambda_N\}$ and divide it into neighborhoods. With individual $x_i \in X$, $B(i) = \{i_1, i_2, \dots, i_T\}$ is then the neighborhood of the i -th weight vector.
- (2) Initialize the species $pop = \{x_1, x_2, \dots, x_N\}$ and calculate its target value $FV_i = F(x_i)$.
- (3) Calculate the ideal point $z = \{z_1, z_2, \dots, z_m\}^T$, where m is the target number and z_i is the optimal value of f_i in the current generation.

Step 2: Update

- (1) Evolution: First, randomly select two individuals x_j and x_k from neighborhood $B(i)$ and use the SBX algorithm to produce a new individual x_d . Then, run the polynomial variant for x_d to obtain the mutant individual y , and then improve it to y' according to the problem constraints.
- (2) Update the ideal point: for $j = 1, 2, \dots, m$, if $f_j(y') < z_j$, then $z_j = f_j(y')$.
- (3) Update the neighborhood: for $j \in B(i)$, if the aggregate function value of y' is not greater than x_j , then $x_j = y'$; in the meantime, update the corresponding target value $FV_j = F(y')$.

Step 3: Stop iteration.

If the stop condition is met, the iteration is stopped and the best solution set is output; otherwise, continue with Step 2.

3. Research on population diversity of MOEA/D algorithm

Based on the above MOEA/D algorithm process, the MOEA/D algorithm is simulated by experiments to the test function, and the experiments show that the algorithm has the problem of prematurity and lack of diversity. Taking the test function DTLZ3 [32] as an example, the MOEA/D algorithm is used to solve its evolution, reproducing 200 generations; the average of the Pareto optimal set of 3 optimization targets is calculated in each generation to reflect the degree of progression of the MOEA/D algorithm, and the change curve of its Pareto optimal set mean value is shown in Fig. 1. This figure shows that, after 40 generations, the mean value of Pareto optimal solution set's reduction trend of the MOEA/D species algorithm decreased significantly, indicating that the evolution speed of the MOEA/D algorithm slowed down.

The reason why the evolution of the MOEA/D algorithm slows down is that the species is lacking in diversity after its proliferation to a certain degree, which leads to the algorithm falling into local optimal solutions. Variance can reflect the diversity of the species: the greater the variance is, the better the diversity of the species is, and vice versa. To this end, the variance mean value in every dimension of each species generation was further studied, as shown in Fig. 2. The significant reduction of the variance mean value of each dimension of the visible species is basically synchronized with the evolutionary

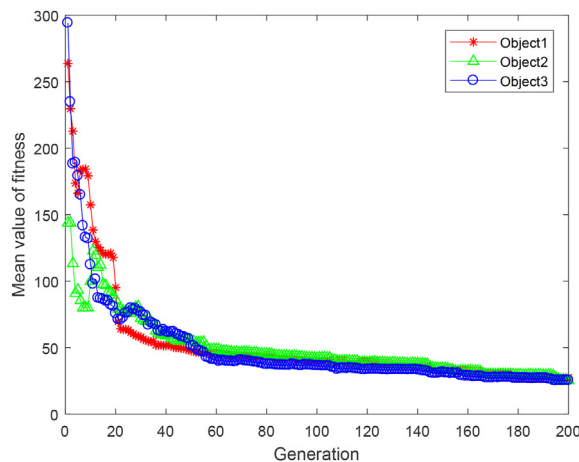


Fig. 1. Mean value curve of Pareto optimal solution set of the MOEA/D algorithm.

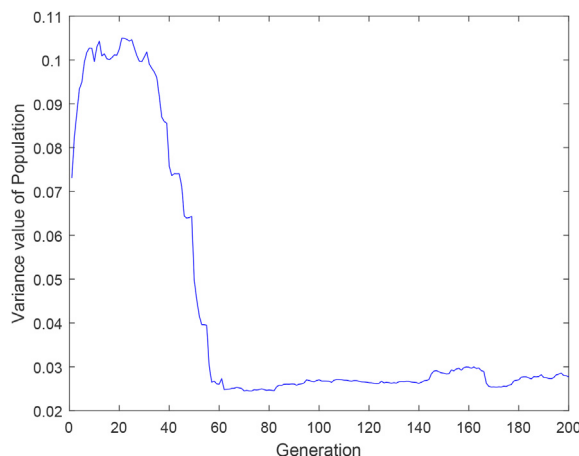


Fig. 2. The variance's mean value curve of the MOEA/D population.

slowdown of the algorithm, which proves that the slowing of the evolution of the MOEA/D algorithm is indeed related to the lack of species diversity.

To solve the problem of slow convergence speed and lack of diversity of the MOEA/D algorithm, this paper has tried three methods for improvement: the first one is to replace the single SBX operator of the original MOEA/D algorithm with the evolutionary strategy of competition between SBX and DE operator; the second one is to replace the fixed mutation probability strategy of the original algorithm with adaptive adjusting strategy of modulation probability; and the last one is a method of double-faced mirror theory boundary optimization which can help to prevent species aggregating at the boundary.

3.1. Evolutionary strategy of competition between SBX and DE operator

The SBX and DE algorithm competitive induction strategy is that in the beginning stage of the algorithm, the DE algorithm is used to perform the crossover operation and record the real-time statistics of each round of evolution of neighbor updates; when the number of neighbor updates significantly decreases, it indicates an evolution slowdown, at which point the SBX algorithm replaces the DE algorithm to perform the crossover operation. These two evolutionary algorithms are used interchangeably with this competitive polling mechanism. Fig. 3 is the Pareto optimal solution set mean value curve of this SBX-DE operator competition evolutionary strategy to improve the MOEA/D algorithm, and Fig. 1 shows that the evolution speed of the species has not stagnated after 40 generations' evolution; instead, it continued to evolve towards PF. Fig. 4 records the change of the variance mean value of the improved MOEA/D algorithm species in each dimension, and the change of the vari-

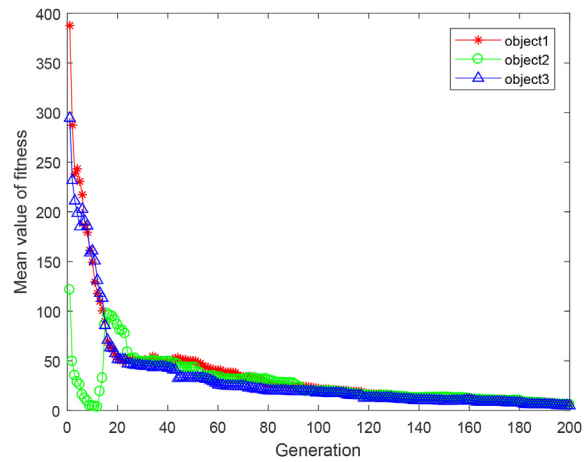


Fig. 3. Pareto optimal solution set mean value curve of the improved MOEA/D algorithm (SBX-DE operator competition strategy).

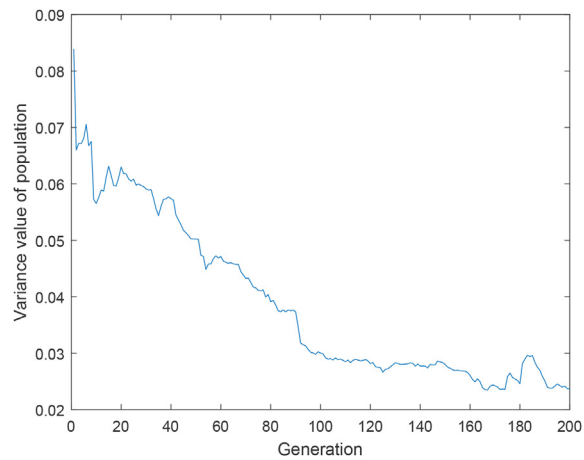


Fig. 4. The curve of the variance mean value of each dimension of the population (SBX-DE operator competition strategy).

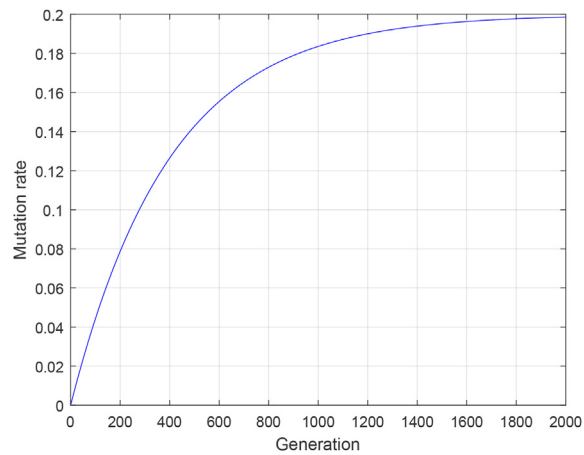


Fig. 5. Adaptive mutation probability adjustment curve.

ance mean value is more stable compared with Fig. 2, and there is no sharp decline, indicating that the SBX-DE operator competition strategy can stabilize the diversity of the species.

3.2. Adaptive adjusting strategy of modulation probability

Another adaptive adjustment strategy of mutation probability is to adjust the individual's probability of variation with the process of evolution, and its adjustment rule is to make the mutation probability change in a negative exponential way with the progressive process; the mutation probability change curve is shown in Fig. 5. The species diversity is better at the beginning stage of evolution, and the probability of variation is lower; with the reduction of species diversity in the evolutionary process, the probability of variation is gradually increased to raise the diversity of the species, but to avoid the problem of reduced convergence caused by the high probability of variation, the probability of variation is increased to a certain value and tends to stabilize. The probability of variation is calculated as follows:

$$Muta_rate = 0.2 * \left(-e^{-\frac{gen}{\chi}} + 1 \right) \quad (3)$$

where gen is the number of generations and χ is the scaling factor. Here, χ is set to 400.

In addition, by recording the number of neighbor updates, the probability of variation is temporarily increased when the algorithm is detected to be stagnant to seek a breakthrough in the stasis. Fig. 6 presents the Pareto optimal solution set mean curve of using the variable probability adaptive adjustment strategy of the improved MOEA/D algorithm; it can be seen that its evolution is faster and has not fallen into an evolutionary stagnation compared with Fig. 1.

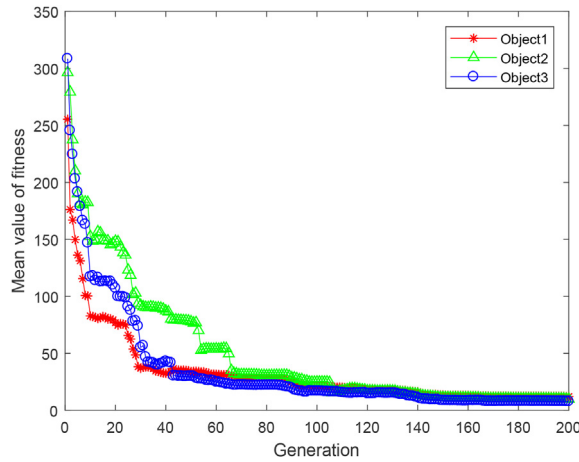


Fig. 6. Pareto optimal solution set mean value curve of the improved MOEA/D algorithm (adaptive adjustment strategy of mutation probability).

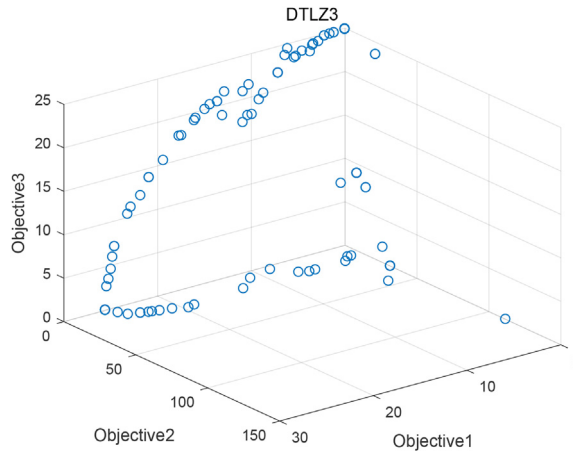


Fig. 7. Pareto optimal solution set distribution map of the MOEA/D algorithm.

3.3. Double-faced mirror theory boundary optimization

Besides, the improper boundary processing leads to uneven distribution of MOEA/D algorithm individuals. In the simulation process of the MOEA/D algorithm, the uneven distribution of Pareto optimal solution sets in the course of evolution is found, which is shown by the gather of solution set near the boundary, while the distribution of other regions is sparse, and it is obviously presented in Fig. 7, and this will directly affect the distribution performance of the PF optimal solution set.

The reason for the uneven distribution of the Pareto optimal solution set of the MOEA/D algorithm is that the algorithm adopts the following method in dealing with boundary problems:

$$y = \begin{cases} X_{\min}, & y < X_{\min} \\ X_{\max}, & y > X_{\max} \\ y, & \text{others} \end{cases} \quad (4)$$

where y is a new evolved individual, $X_{\min}$ is the lower boundary limit, and $X_{\max}$ is its boundary upper limit, which will allow all of the out-bound individuals to be projected to the boundary aggregation.

To this end, a double-faced mirror reflection boundary processing algorithm is proposed, as is shown in Fig. 8, with a boundary $[X_{\min}, X_{\max}]$ and y as a cross-boundary point, where the upper and lower boundaries are two mirrors; y is the beam propagated inside, the size of y indicates light intensity, and the middle medium has light loss. This beam of light will eventually run out at some point y' within the boundary due to media loss after multiple specula reflections. If there are several

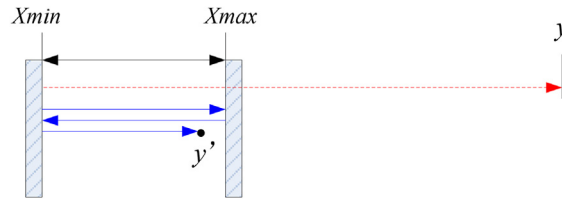


Fig. 8. Double-faced mirror reflection schematic.

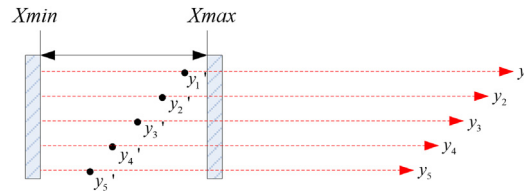


Fig. 9. Double-faced mirror reflection projection effect.

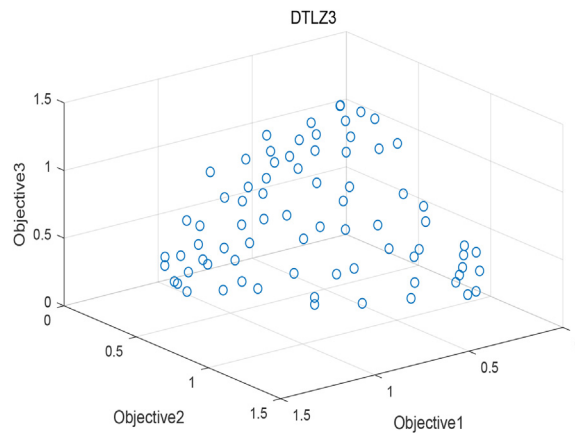


Fig. 10. Pareto optimal solution set distribution map of the improved MOEA/D algorithm (border problems are handled by the double-faced mirror).

beams of continuous strength values, the final dissipation points of these beams through double-faced mirror reflection sit through the boundary area will be evenly distributed within the boundary area, as is shown in Fig. 9. From this, it can be seen that the principle of double-faced mirror reflection can solve the problem of uneven distribution of boundary processing.

The formula for dealing with boundary problems of the double-faced mirror reflection principle is as follows:

$$y' = \begin{cases} X_{\max} - \text{mod}(y - X_{\max}, X_{\max} - X_{\min}), & y > X_{\max} \\ X_{\min} + \text{mod}(X_{\min} - y, X_{\max} - X_{\min}), & y < X_{\min} \end{cases} \quad (5)$$

Applying the principle of double-faced mirror reflection to the boundary processing of the MOEA/D algorithm can effectively solve the problem of uneven distribution of its Pareto optimal solution set, as is shown in Fig. 10.

4. AES-MOEA/D algorithm

The research in the previous section shows that the above three improvements can significantly increase the diversity of the MOEA/D algorithm population. Based on the research, an improved MOEA/D algorithm, Adaptive Evolutionary Strategy MOEA/D (AES-MOEA/D), is proposed in this paper. The AES-MOEA/D algorithm is intended to study the performance characteristics of the improved MOEA/D algorithm in the condition of synthesizing the above three improvements to fully maintain the population diversity.

The AES-MOEA/D algorithm adopts the basic framework of the MOEA/D algorithm and increases the mutation probability calculation and neighbor update frequency statistics recording in the course of evolution to identify and adapt to the degree of species evolution. In the course of evolution, the MOEA/D algorithm evolutionary operator randomly selects individuals in the neighborhood to generate a new individual y' using cross-variation, and if the aggregate function value of the new individual y' is better than the φ_i individual solution in the corresponding neighborhood, the new individual y' will update the φ_i individual solution. Therefore, this neighborhood update frequency φ_i is a good representation of the evolutionary state of the subproblem corresponding to the neighborhood.[33] When the neighborhood update frequency φ_i is above a certain threshold, it indicates that the MOEA/D algorithm is in the search phase, and when φ_i is below a certain threshold, it indicates that the MOEA/D algorithm has evolved at a rate of change, or even stagnated. The AES-MOEA/D algorithm also uses the double-faced mirror theory to deal with the cross-boundary problem of cross-mutated individuals.

The implementation steps of AES-MOEA/D algorithm are as follows:

Step 1: Initialization

Step 1.1 Initialize the weight vector $\lambda = \{\lambda_1, \lambda_2, \dots, \lambda_N\}$ and divide the neighborhood. With individual $x_i \in X$, $B(i) = \{i_1, i_2, \dots, i_T\}$ is the i th neighborhood of the weight vector.

Step 1.2 Initialize the species $\text{pop} = \{x_1, x_2, \dots, x_N\}$ and calculate its target value $FV_i = F(x_i)$.

Step 1.3 Calculate the ideal point $z = \{z_1, z_2, \dots, z_m\}^T$ and set m as the number of targets; z_i is the optimal value of f_i in the current generation

Step 2: Evolution

Step 2.1 Calculate the probability of variation ξ according to formula (3), where the parameter gen is the current iteration

Step 2.2 According to the evolutionary strategy, two individuals x_j and x_k are randomly selected from the neighborhood $B(i)$, and the SBX or DE operator is selected for cross-evolution based on the population evolution speed, resulting in a new individual x_d .

Step 2.3 Based on Step 2.1, the probability of variation ξ is obtained. The polynomial variation of x_d obtains individual y , which is finally improved to y' according to the problem constraint. If there is a cross-border problem, use the double-faced mirror theory to calculate it using formula (5).

Step 2.4 Update the ideal point: for $j = 1, 2, \dots, m$; if $f_j(y') < z_j$, then $z_j = f_j(y')$.

Step 3: Update the neighborhood relationship

Step 3.1 Update the neighborhood: for $j \in B(i)$, if y' aggregate function value is not greater than x_j , then $x_j = y'$, and the corresponding target value $FV_j = F(y')$ is updated.

Step 3.2 Count the neighbor update frequency of this iteration and update the neighbor update frequency that records the last five iterations.

Step 4: Adjust the evolutionary strategy

If the last five neighbor update frequencies are below the threshold β , the cross-evolutionary algorithm needs to be replaced, and the variation frequency is temporarily increased. Otherwise, the last crossover algorithm will continue to be used, and the mutation probability will also be the result that was calculated in Step 2.1.

Step 5: Stop condition judgment

Determine whether to stop the iteration: If the stop condition is met, the iteration is stopped and the best solution set is output; otherwise, continue with Step 2.

Then the framework of AES-MOEA/D is as follow:

Algorithm 1. (Framework of the AES-MOEA/D.).

Inputs: λ -weight vector of MOPs, B -neighborhood, X - individual initial position, z - reference point
Outputs: FV - object value of X
for $gen = 1$ to $Maxgen$ **do**
 Calculate the probability of variation ζ : $\zeta \leftarrow 0.2 * \left(-e^{-\frac{gen}{\lambda}} + 1\right)$
 for $i = 1$ to N **do** //Where N is the number of individuals of the population.
 if $Mod = 0$ **then**
 Randomly select three individuals from $B(i)$.
 Generate difference individual by using DE operator: $y \leftarrow x_1 + CF * (x_2 - x_3)$
 else
 Randomly select two individuals from $B(i)$.
 Generate crossover individual by using SBX operator: $y \leftarrow \frac{1}{2} * ((1 - \beta_i) * x_1 + (1 + \beta_i) * x_2)$
 end if
 Generate a random number r within $[0,1]$.
 if $r < \zeta$ **then**
 Polynomial variation: $y \leftarrow y + \beta_i$
 end if
 Check bound by using double-faced mirror theory according to formula (5).
 Calculate object value of y : $fit_y \leftarrow F(y)$.
 for $j = 1$ to M **then** //Where M is the object dimension.
 if $fit_y(j) < z(j)$ **then**
 $z(j) \leftarrow fit_y(j)$ // Chebyshev method
 end if
 end for
 Update neighborhood: $[X, FV] \leftarrow updateNeigh(X, FV, B(i), y, fit_y)$.
 end for
 Statistics update number of the individuals: $m \leftarrow cal.update(X)$.
 if $m < \beta$ **then**
 $Mod \leftarrow (Mod)$
 end if
end for

5. Experimental analysis

To detect the performance of the AES-MOEA/D algorithm, this paper selects four classic MOP evolutionary algorithms, MOEA/D, NSGA-II, SPEA-II and PESA-II as comparison algorithms of AES-MOEA/D and conducts optimization simulation comparison experiments of these algorithms on test functions. To analyze the performance of the five algorithms intuitively, this paper first analyzes the optimization and comparison experiments of the five algorithms on the test function DTLZ3 (set 12 decision variables and 3 target numbers), and the approximate PF image of them is shown in Fig. 11 after 1000 generations of evolution.

As seen from Fig. 11, the PF approximation set of the SPEA-II algorithm is distributed uniformly and has the best distribution, but its convergence is the worst; the PESA-II algorithm is also well distributed, but the convergence is also worse; and the convergence performances of the NSGA-II algorithm and the AES-MOEA/D algorithm are better. However, the distribution of the NSGA-II algorithm is poor, and the AES-MOEA/D algorithm is improved compared with the MOEA/D algorithm both in convergence and distribution. It can be seen that the overall performance of AES-MOEA/D algorithm is the best of the five algorithms on the test function DTLZ3.

To roundly compare the overall performance of the five algorithms, ZDT1, ZDT2, ZDT3, ZDT4, ZDT6, DTLZ2, DTLZ3, DTLZ6, and DTLZ10 were selected as the optimization test objects [34,35], the maximum spread (MS), the inverted generation distance (IGD), and hyper volume (HV) were selected as the evaluation basis, and the above five algorithms were evaluated from the three aspects: coverage, convergence and comprehensiveness.

The MS indicator is used to describe the coverage performance of the algorithm; it represents the degree of coverage [36] of the real PF obtained by the PF, and its definition is shown in formula (6); the higher the MS value, the larger the area covered by the approximate solution set S , and the greater the diversity.

$$MS = \sqrt{\frac{1}{m} \sum_{i=1}^m \left(\frac{\min(f_i^{\max}, F_i^{\max}) - \max(f_i^{\min}, F_i^{\min})}{F_i^{\max} - F_i^{\min}} \right)^2} \quad (6)$$

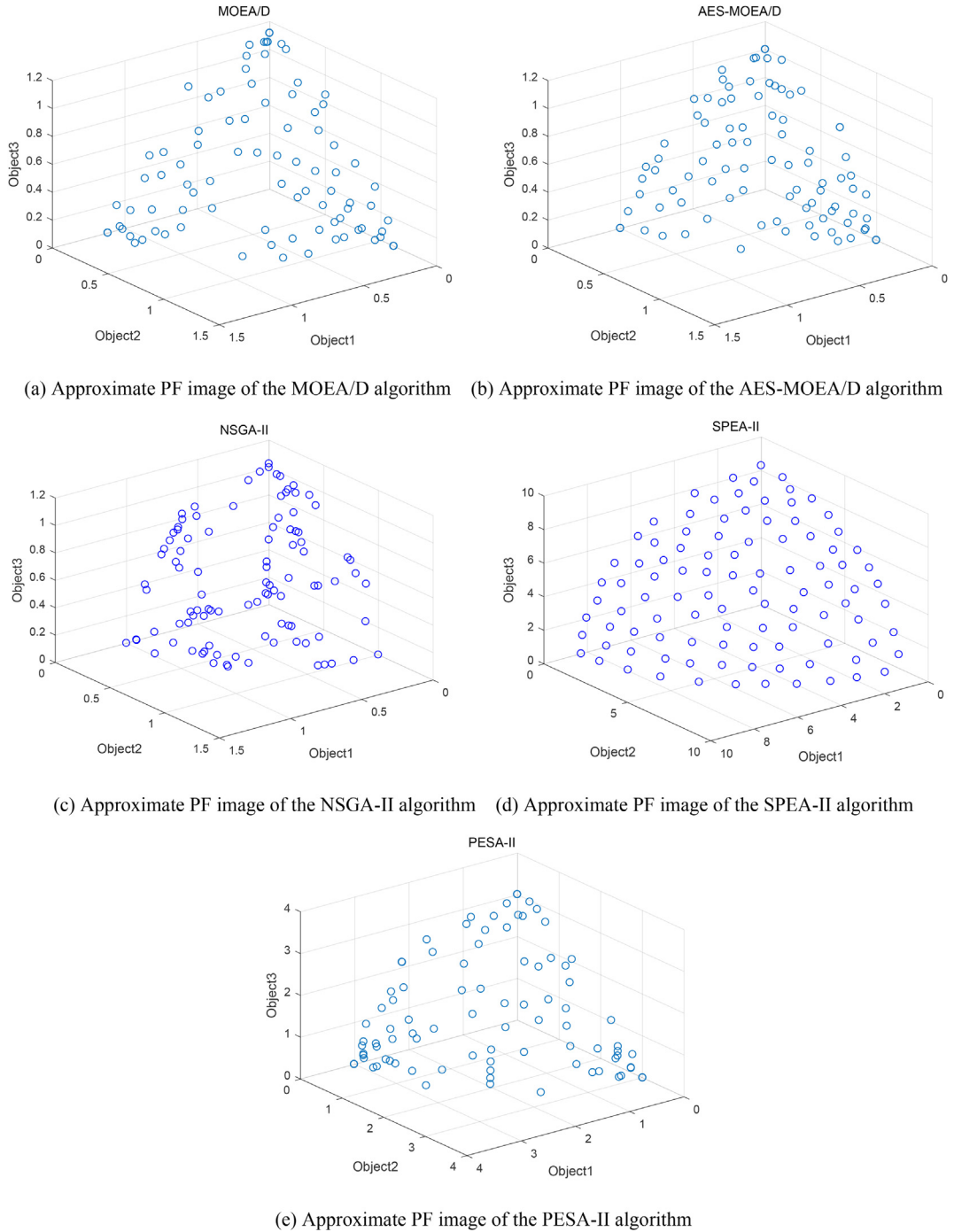


Fig. 11. DTLZ3 approximate PF image of the five algorithms.

To ensure equity, the experiment uniformly set population as 100 and 1000 iterations of each algorithm. Each algorithm is independently calculated 5 times in each test function to take the mean value. The experiment obtained MS indicators of five algorithms in 10 test functions, which is shown in Table 1. From Table 1, it can be seen that the coverage performance of the AES-MOEA/D algorithm is close to that of the NSGA-II, SPEA-II and MOEA/D algorithms and is superior to that of the PESA-II algorithm.

The IGD indicator [37] is the mean value of the minimum distance between the individual of real PF and the approximate solution sets obtained by the algorithm, which can reflect the convergence of the algorithm. Set P^* as a solution set that is

Table 1

MS indicators of five algorithms in 10 test functions.

Test function		NSGA-II		SPEA-II		PESA-II		MOEA/D		AES-MOEA/D
ZDT1	mean	1.00E+00	≈	1.00E+00	≈	9.61E−01	+	1.00E+00	≈	9.98E−01
	var	7.16E−26		1.38E−18		7.63E−04		6.77E−09		5.17E−07
ZDT2	mean	1.00E+00	≈	1.00E+00	≈	9.97E−01	≈	1.00E+00	≈	1.00E+00
	var	0.00E+00		0.00E+00		1.95E−05		7.30E−09		2.00E−01
ZDT3	mean	1.00E+00	≈	1.00E+00	≈	9.94E−01	+	9.58E−01	+	9.98E−01
	var	2.19E−09		2.42E−19		3.88E−05		8.38E−03		6.56E−08
ZDT4	mean	8.23E−01	+	9.53E−01	+	9.86E−01	+	1.00E+00	−	9.96E−01
	var	7.40E−04		1.10E−02		2.82E−04		3.23E−07		1.68E−05
ZDT6	mean	1.00E+00	≈	1.00E+00	≈	9.97E−01	≈	1.00E+00	≈	9.99E−01
	var	0.00E+00		9.72E−10		3.35E−05		3.33E−15		3.32E−09
DTLZ1	mean	1.18E−02	+	4.04E−02	+	4.04E−02	+	1.00E+00	≈	1.00E+00
	var	0.00E+00		0.00E+00		0.00E+00		0.00E+00		4.68E−11
DTLZ2	mean	1.00E+00	≈	1.00E+00	≈	1.00E+00	≈	1.00E+00	≈	1.00E+00
	var	0.00E+00		0.00E+00		0.00E+00		0.00E+00		7.47E−12
DTLZ3	mean	1.00E+00	≈	1.00E+00	≈	1.00E+00	≈	1.00E+00	≈	1.00E+00
	var	0.00E+00		0.00E+00		0.00E+00		0.00E+00		2.43E−09
DTLZ5	mean	6.57E−01	−	7.37E−01	−	5.84E−01	≈	5.03E−01	+	5.86E−01
	var	2.51E−04		0.00E+00		5.84E−01		3.34E−02		2.11E−05
DTLZ6	mean	1.00E+00	≈	1.00E+00	≈	1.00E+00	≈	1.00E+00		1.00E+00
	var	9.56E−07		0.00E+00		8.17E−03		2.28E−04		6.50E−05

Notes: '+' means that AES-MOEA/D performs better than that of the comparing algorithm.

'≈' means that AES-MOEA/D performs similarly with that of the comparing algorithm.

'−' means that AES-MOEA/D performs worse than that of the comparing algorithm.

evenly sampled on real PF, S is the PF approximation solution set obtained by the multi-objective evolutionary algorithm, and the IGD definition is shown in equation (7):

$$IGD(S, P^*) = \frac{\sum_{x \in P^*} dist(x, S)}{|P^*|} \quad (7)$$

In above equation, $dist(x, S)$ represents the Euclidean distance between the individual $x \in P^*$ and the nearest individual of S , and $|P^*|$ is the base point of set P^* . The smaller the IGD value is, the better the convergence and diversity of S are, and the closer it can approach the entire PF. The IGD indicators of the five algorithms are obtained in the experiment as is shown in Table 2. From Table 2, it can be seen that the IGD index of the AES-MOEA/D algorithm is better than that of the other 4 comparison algorithms, and it only performs worse on the test functions ZDT1, ZDT3 and DTLZ5.

Table 2

IGD indicators of five algorithms in 10 test functions.

Test function		NSGA-II		SPEA-II		PESA-II		MOEA/D		AES-MOEA/D
ZDT1	mean	4.78E−01	−	4.89E−01	−	4.82E−01	−	5.22E−01	≈	5.21E−01
	var	5.34E−04		3.14E−05		2.28E−03		5.07E−06		4.94E−06
ZDT2	mean	6.53E−01	+	6.35E−01	+	6.21E−01	+	6.54E−01	+	2.10E−01
	var	3.48E−04		3.62E−06		1.05E−03		1.73E−05		6.05E−02
ZDT3	mean	6.58E−01	−	6.44E−01	−	6.47E−01	−	9.36E−01	−	9.54E−01
	var	1.71E−03		9.47E−05		1.51E−03		1.53E−03		9.46E−06
ZDT4	mean	2.06E+00	+	5.93E−01	+	4.58E−01	−	4.99E−01	≈	4.98E−01
	var	6.22E−03		5.43E−02		2.63E−03		1.78E−06		2.64E−06
ZDT6	mean	4.88E−01	+	5.23E−01	+	5.53E−01	+	5.81E−01	+	4.82E−01
	var	2.84E−04		2.77E−05		7.84E−03		1.65E−05		9.45E−06
DTLZ1	mean	1.49E+02	+	1.80E+02	+	1.80E+02	+	4.35E−01	+	4.29E−01
	var	9.68E+01		0.00E+00		0.00E+00		9.15E−05		2.06E−04
DTLZ2	mean	8.27E−01	+	8.01E−01	+	8.38E−01	+	7.56E−01	+	7.27E−01
	var	5.86E−04		1.54E−32		1.92E−03		2.50E−03		3.19E−03
DTLZ3	mean	9.23E+01	+	1.04E+01	+	7.04E+02	+	1.52E+00	+	6.82E−01
	var	1.54E+03		0.00E+00		3.63E+00		1.15E+00		3.87E−04
DTLZ5	mean	2.78E+00	−	2.89E+00	−	1.70E+00	−	3.08E+00	−	3.12E+00
	var	3.25E−04		0.00E+00		1.72E−03		1.13E−02		2.21E−03
DTLZ6	mean	4.46E+00	+	4.47E+00	+	4.72E+00	+	3.87E+00	≈	3.88E+00
	var	9.91E−04		0.00E+00		1.84E−02		1.44E−03		7.77E−03

Notes: '+' means that AES-MOEA/D performs better than that of the comparing algorithm.

'≈' means that AES-MOEA/D performs similarly with that of the comparing algorithm.

'−' means that AES-MOEA/D performs worse than that of the comparing algorithm.

HV [38] is a measure of comprehensive performance that refers to a given set of preset reference points $r^* = (r_1^*, r_2^*, \dots, r_m^*)$ distributed in the target space and a set of algorithmic PF approximate solutions S , satisfying that r^* is dominated by all of the solutions in S . HV measures the volume size of the target space bounded by r^* and dominated by S , as is defined in equation (8):

$$HV(S) = VOL(\cup_{x \in S} [f_1(x), r_1^*] \times \dots [f_m(x), r_m^*]) \quad (8)$$

In the above equation, $VOL(\cdot)$ indicates the Lebesgue measure [39]. The higher the HV value is, the more S is approximated to the entire PF. The HV indicator of five algorithms is shown in Table 3. From Table 3, we can see that the HV indicator of the AES-MOEA/D algorithm is better than that of the other 4 comparison algorithms, and it only performs worse on the test function DTLZ1.

In order to further study the performance of the AES-MOEA/D algorithm in optimizing many-objective problems (usually the number of targets is greater than 3), this paper sets up a group of 10-objective test experiments: the test function DTLZ1 is used as the test object, its target number is set as 10, and the number of decision variables is set as 12, and then the performances of the four algorithms MOEA/D, AES-MOEA/D, SPEA-II, and NSGA-II are tested in this problem. To ensure the fairness of the comparison experiment, in this experiment, the population number of the four algorithms is set as 220, the number of iterations is 100 times, and each algorithm is independently run 50 times to obtain 50 groups of experimental data for each algorithm.

In the study of many-objective optimization problems, parallel coordinates are a commonly used method for high-dimensional geometry and multivariate data visualization [40]. In this paper, the non-dominated solution with the smallest IGD value in the DTLZ1 problem is selected as the evaluation index, and the parallel coordinate plot of its 10 objects are drawn as Fig. 12.

It can be seen from Fig. 12 that the mean values of the optimization target values of the MOEA/D and AES-MOEA/D algorithms are smaller than those of the SPEA-II and NSGA-II algorithms, which means that the convergence of MOEA/D and AES-MOEA/D algorithms is better than that of SPEA-II and NSGA-II algorithms in optimizing many-objective problems, and the convergence of the AES-MOEA/D algorithm is slightly better than that of the MOEA/D algorithm. On the other hand, in the 10 target dimensions of DTLZ1, each set of objective values of the MOEA/D and AES-MOEA/D algorithms has smaller fluctuations (expressed as a densely distributed area in the middle area of the ordinate of the image), while the SPEA-II and NSGA-II algorithms have larger fluctuations in each set of objective values (represented by no obvious densely distributed areas in the ordinate direction of the image). It indicates that the MOEA/D and AES-MOEA/D algorithms have a relatively stable distribution of the objective values of each target dimension in many-objective problems, that is, the differences between the optimization targets are relatively small, which is beneficial to the balanced decision-making of the many-objective problems. In general, the AES-MOEA/D algorithm is superior to the three comparison algorithms: MOEA/D, SPEA-II, and NSGA-II, in terms of convergence and balance of each objective in the 10-target DTLZ1 optimization problem.

Table 3
HV indicator of five algorithms in 10 test functions.

Test function		NSGA-II		SPEA-II		PESA-II		MOEA/D		AES-MOEA/D
ZDT1	mean	9.17E+03	≈	9.17E+03	≈	9.19E+03	≈	9.19E+03	≈	9.19E+03
	var	2.22E+01		3.45E−01		6.62E+01		3.55E−01		4.40E−01
ZDT2	mean	8.86E+03	+	8.86E+03	+	8.85E+03	+	8.87E+03	+	8.97E+03
	var	4.42E+00		3.63E−01		2.14E+01		3.24E−01		3.25E+03
ZDT3	mean	9.45E+03	+	9.46E+03	+	9.48E+03	+	9.58E+03	≈	9.58E+03
	var	2.91E+01		6.06E+01		5.98E+02		9.75E+00		5.93E+00
ZDT4	mean	7.30E+03	+	9.02E+03	+	9.18E+03	≈	9.19E+03	≈	9.19E+03
	var	7.06E+03		1.12E+05		5.17E+02		7.87E+00		1.12E+00
ZDT6	mean	8.84E+03	≈	8.85E+03	≈	8.77E+03	+	8.86E+03	≈	8.86E+03
	var	3.87E+01		6.03E−01		1.21E+04		1.04E+00		4.65E−01
DTLZ1	mean	1.39E+07	−	2.61E+07	−	2.61E+07	−	1.13E+04	≈	1.13E+04
	var	6.25E+12		0.00E+00		0.00E+00		9.97E+00		1.21E+00
DTLZ2	mean	8.57E+04	+	8.59E+04	+	8.55E+04	+	8.68E+04	≈	8.69E+04
	var	1.03E+04		0.00E+00		1.17E+04		3.17E+04		1.06E+04
DTLZ3	mean	0.00E+00	+	9.07E+03	+	0.00E+00	+	7.55E+04	+	8.63E+04
	var	0.00E+00		0.00E+00		0.00E+00		1.81E+08		1.62E+04
DTLZ5	mean	7.95E+04	+	8.10E+04	+	8.28E+04	+	8.47E+04	−	8.44E+04
	var	3.10E+05		0.00E+00		6.65E+05		1.70E+06		1.22E+05
DTLZ6	mean	4.99E+04	+	4.98E+04	+	4.82E+04	+	5.40E+04	≈	5.38E+04
	var	2.75E+04		0.00E+00		5.71E+05		5.32E+04		3.10E+05

Notes: '+' means that AES-MOEA/D performs better than that of the comparing algorithm.

'≈' means that AES-MOEA/D performs similarly with that of the comparing algorithm.

'−' means that AES-MOEA/D performs worse than that of the comparing algorithm.

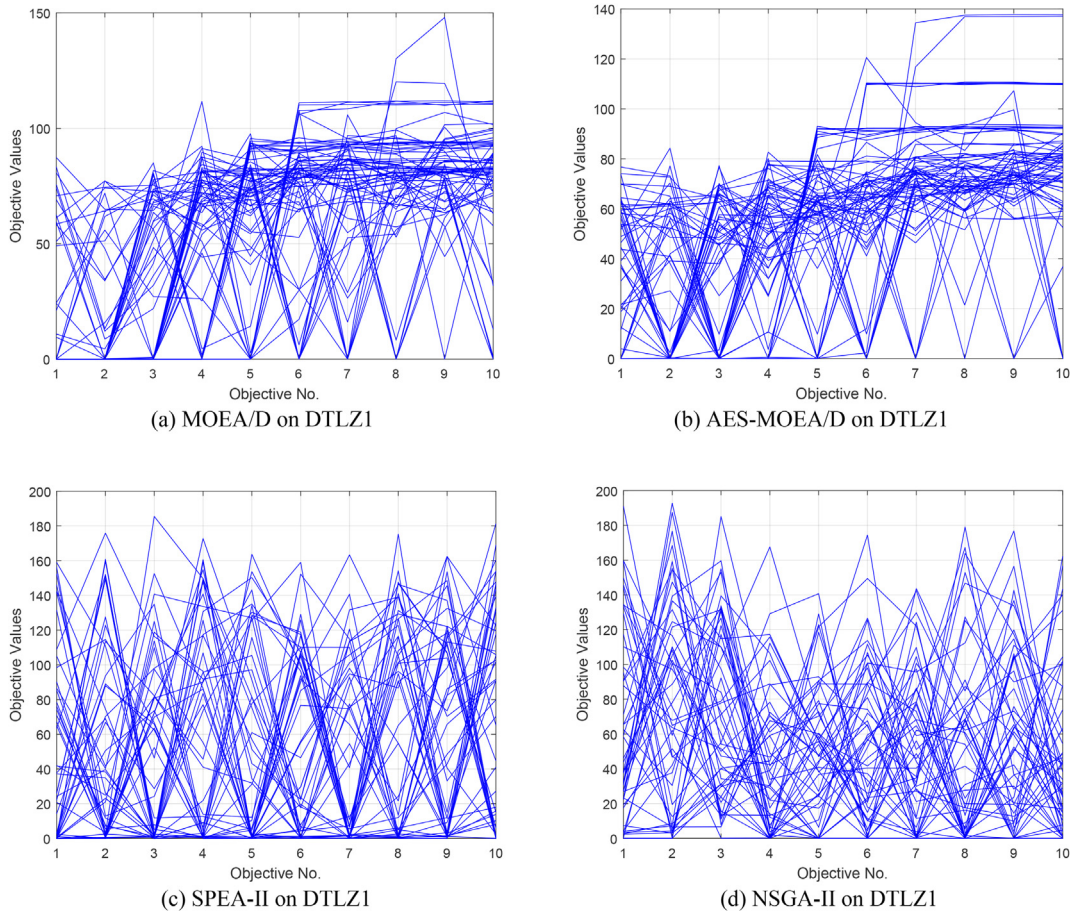


Fig. 12. The non-dominated solutions with the lowest IGD metric values obtained by each algorithm among 50 runs in the objective space on 10-objective DTLZ1 problem with 12 decision variables.

6. Conclusion

Aimed at the problems of slow convergence and lack of population diversity in the late evolution stage of MOEA/D algorithm, this paper systematically studies the strategies of maintaining population diversity of MOEA/D algorithm, and proposes three improvement measures: evolutionary strategy of competition between SBX and DE operator, adaptive adjusting strategy of modulation probability and double-faced mirror theory boundary optimization. Then the effectiveness of these measures is verified. On this basis, the AES-MOEA/D algorithm is proposed, and the performance characteristics of the improved MOEA/D algorithm in the condition of fully maintaining the diversity of the population by using the above three improvement measures are studied. With the comparison simulation experiments of the MOEA/D, NSGA-II, SPEA-II and PESA-II algorithms in 10 test functions, we conclude that the AES-MOEA/D algorithm is equivalent in coverage performance to that of NSGA-II, SPEA-II and MOEA/D algorithms and better than that of PESA-II algorithm. Further, its overall performance is better than that of the other four comparison algorithms in convergence and comprehensiveness. In addition, the AES-MOEA/D algorithm's convergence and equilibrium are also superior to those of the three comparison algorithms in the optimization experiments of many-objective problems. It is indicated that improving the population diversity of the MOEA/D algorithm in the late evolutionary stage is conducive to improving its comprehensive performance.

CRedit authorship contribution statement

Wang Wen-xiang: Software, Validation, Writing - original draft, Writing - review & editing. **Li Kang-shun:** Methodology, Supervision, Project administration. **Tao Xing-zhen:** Resources, Data curation. **Gu Fa-hui:** Funding acquisition, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by Key-Area Research and Development Program of Guangdong Province, China with the Grant No. 2018YFC0831100, the National Natural Science Foundation of China, China with the Grant No. 61773296, the National Natural Science Foundation Youth Fund Project of China, China under Grant No. 61703170, Foreign Science and Technology Cooperation Program of Guangzhou, China with the grant No. 201907010021, the Opening Project of Guangdong Key Laboratory of Big Data Analysis and Processing, China with No. 201901, the Major Science and Technology Project in Dongguan, China with No. 2018215121005. Key R&D Program of Guangdong Province, China with No. 2019B020219003, Foreign Science and Technology Cooperation Program of Huangpu District of Guangzhou, China with No. 2018GH09. And is supported by the Science and Technology Project of Ganzhou, China with No. GSKF201850 and the Science and Technology Project of Education Department of Jiangxi Province, China with GJJ181265.

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